

The Impact of AI Agent Personality in Virtual Debates

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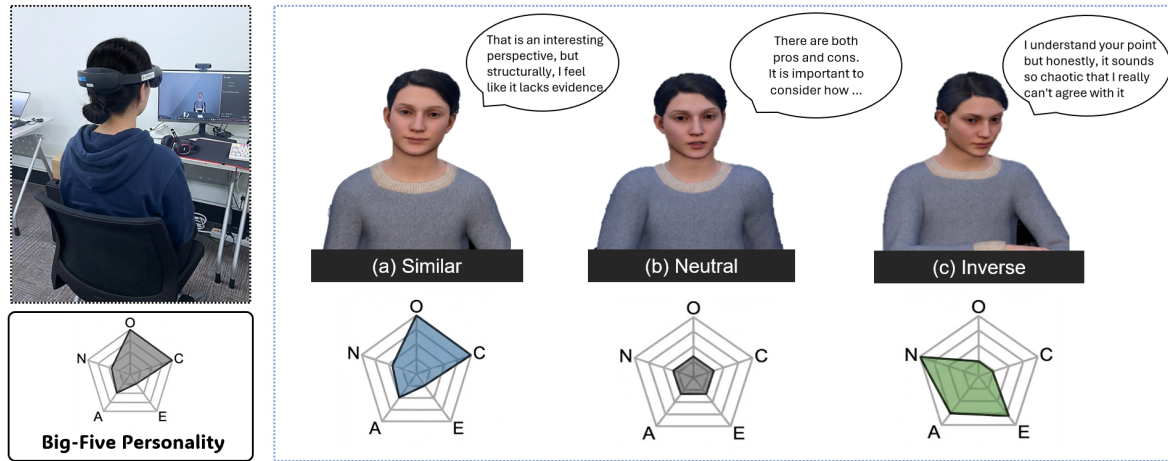


Figure 1: We compared three personality-matching conditions in a debate scenario including agreement and disagreement debating scenarios. The experimental conditions were defined based on the participants' Big Five personality traits and were expressed both verbal and nonverbal behaviors: (a) Similar (matching traits), (b) Neutral (no expression), and (c) Inverse (opposing traits).

ABSTRACT

Recent advances in large language models have enabled Embodied Conversational Agents (ECAs) to function as social actors capable of rich, multimodal interaction. In this work, we investigate how adapting an agent's personality to the user influences social connection and persuasion in debate contexts. We conducted a user study comparing three personality conditions in VR debate interactions: Similar (matching traits), Neutral, and Inverse (opposing traits). Our results showed that similar personality consistently enhances social presence and social connection, highlighting the importance of personality-adaptive ECAs in social interactions.

Index Terms: Virtual agents, Personality, Persuasion

1 INTRODUCTION

Recent advances in Large Language Models (LLMs) have transformed Embodied Conversational Agents (ECAs) into capable social actors [7]. As ECAs become visually embodied and engage in complex social interactions, understanding how they should express social cues becomes a critical design challenge. Among various cues, personality expression has been shown to facilitate rapport and enhance user experience in human-agent interaction [10, 3].

Despite the importance of personality expression in ECAs, the optimal strategy for adapting an agent's personality to the user remains unclear. To guide this adaptation strategy, we draw on two theoretical approaches that form the basis of our experimental conditions. One approach is grounded in the similarity-attraction hypothesis [2], which suggests that people tend to respond more positively to interaction partners who resemble themselves. Conversely,

the other approach highlights the potential of complementarity [11], indicating that opposing traits (e.g., dominance paired with submissiveness) can be socially impactful by creating a functional balance.

In this paper, we investigate how personality alignment between users and ECAs influences social connection and persuasion in debate interactions. We design an ECA whose personality is adapted to each user's Big Five profile and expressed through both verbal and nonverbal behaviors. Specifically, we compare three experimental conditions—Similar, Neutral, and Inverse—across two debate tasks: agreement, in which the agent and user argue on the same side, and disagreement, in which they take opposing positions.

2 METHODOLOGY

2.1 ECA Design

We implemented an ECA capable of expressing distinct personality traits grounded in the Big Five model [9], which includes Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. The ECA was created using a MetaHuman character¹, and its personality was expressed through both verbal and nonverbal behaviors. **Verbal output** was generated using GPT-4o² with a two-stage prompting strategy: the agent was first assigned a background persona defined by Big Five traits, and then instructed on how these traits should be linguistically expressed through word choice and sentence structure [8]. **Nonverbal expressions** were conveyed through listening behaviors, speaking gestures, and facial expressions. Speaking gestures were generated following the LMA mappings defined in PERFORM [4], and facial expressions were controlled using the Convai emotion framework³, adjusting the intensity of five basic emotions to reflect the agent's personality profile [10].

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¹<https://metahuman.unrealengine.com/>

²<https://openai.com/index/hello-gpt-4o/>

³<https://convai.com/blog/realistic-face-animations-ai-characters-convai>

2.2 Experimental Design

We employed 3 (**Personality**: *Similar*, *Neutral*, *Inverse*) $\times$ 2 (**Conversation Type**: *Agreement*, *Disagreement*) mixed factorial design, with Personality as a within-subjects factor and Conversation Type as a between-subjects factor. Participants interacted with ECAs under three **Personality** conditions derived from their Big Five profiles (Figure 1): (1) *Similar*, matching the participant's traits ($x_s = x$); (2) *Neutral*, using neutral language and non-expressive behaviors; and (3) *Inverse*, exhibiting opposite traits ($x_i = O^-, C^-, E^+, A^-, N^+$). For **Conversation Type**, participants were assigned to either an *Agreement* or *Disagreement*, in which the ECA aligned with or opposed the participant's initial stance.

Participants We recruited 72 participants from a local university. The Agreement group ($N = 36$) included 22 males and 14 females, with ages ranging from 18 to 42 ($M = 24.9$, $SD = 6.3$). The Disagreement group ($N = 36$) included 12 males and 24 females, with ages ranging from 20 to 46 ($M = 23.9$, $SD = 4.7$).

Procedure The study was approved by the Institutional Review Board (IRB) of the host institution. It began with a pre-survey to collect demographic information and assess participants' personality using Big Five Inventory [5]. Additionally, participants rated ten pre-defined debate topics for their stance, confidence, and knowledge level. We then selected three similarly rated topics per participant and randomly assigned to the three conditions. For the main experiment, participants wore a Meta Quest Pro HMD and entered the virtual room. Interactions took place under three counterbalanced conditions using a Latin Square design. Each session adhered to turn-taking protocol, followed by a post-trial survey.

3 RESULTS AND DISCUSSION

We used set of questionnaire measuring perceived similarity, social presence [6], Inclusion of Others in the Self (IOS) [1], and social influence [7]. We conducted a mixed ANOVA with Aligned Rank Transform (ART), followed by ART-C post-hoc tests with Bonferroni correction. The significance level was set at $\alpha = .05$.

For **Perceived Similarity**, a significant main effect of Personality Condition was found ($F(2, 140) = 14.98$, $p < .001$), with the *Similar* agent rated higher than both *Neutral* ($p = .016$) and *Inverse* ($p < .001$), and *Neutral* rated higher than *Inverse* ($p = .02$). Similarly, we found a significant main effect of Personality Condition on **Social Presence** ($F(2, 140) = 5.32$, $p = .006$), with higher ratings in the *Similar* than in the *Inverse* condition ($t = -3.92$, $p < .001$), while no other pairwise differences were significant. **IOS** showed significant main effects of Personality Condition ($F(2, 140) = 8.79$, $p < .001$) and Conversation Type ($F(1, 70) = 8.93$, $p < .001$). Post-hoc tests showed higher scores in the *Similar* condition compared to the *Neutral* ($t = -2.38$, $p = .048$) and *Inverse* ($t = -4.18$, $p < .001$) conditions, with no difference between the latter two. Finally, **Social Influence** ratings showed a significant main effect of Personality Condition ($F(2, 140) = 7.73$, $p < .001$), with greater persuasion in the *Similar* than in the *Inverse* condition ($t = 2.651$, $p = .024$). There was no significant main effect of Conversation Type ($p = .702$) and no significant interaction ($p = .777$).

Post-task feedback indicated that participants could distinguish personality conditions, primarily through verbal behaviors. About 75% perceived differences mainly in word choice, tone, and argumentation style, likely due to the debate-oriented setting.

4 CONCLUSION

In this paper, we explored how personality matching between an ECA and a user influences social perception in a VR debate. We designed a multimodal ECA that expresses diverse personality styles based on individual Big Five traits. Personality was modeled as fixed and persuasion was assessed only through self-report, limiting adaptivity and long-term validity. Future work will examine adaptive personality systems and behavioral or longitudinal outcomes.

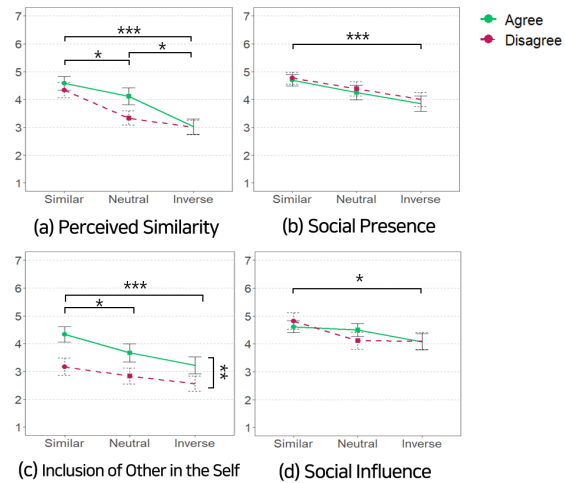


Figure 2: Results of subjective measures (* $p < .05$; *** $p < .001$)

ACKNOWLEDGMENTS

This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) under the Artificial Intelligence Convergence Innovation Human Resources Development (IITP-2023-RS-2023-00256629) and Innovative Human Resource Development for Local Intellectualization program (IITP-2026-RS-2022-00156287) grant funded by the Korea government(MSIT).

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